FC7xxx Port User Manual

Rev A0

Revision History

Revision	Date	Changes
A0	2025/01/14	Initial release

Table of Contents

Chapter 1 PORT Introduction

1.1 Requirement Tracing

The design of this module follows the specifications of the PORT driver specified in AUTOSAR Classic Platform Release 4.3.1. For detailed requirements, refer to the *AUTOSAR_SWS_PortDriver*.

1.2 Hardware Summary

This module provides the service for initializing the whole PORT structure of the microcontroller. Many ports and port pins can be assigned to various functionalities, e.g.

- General purpose I/O
- ADC
- SPI
- FTU
- UART
- FLEXCAN
- LIN

For this reason, there is an overall configuration and initialization of this port structure. Port initialization data shall be written to each port as efficiently as possible. This PORT driver module completes the overall configuration and initialization of the port structure which is used in the DIO driver module. Therefore, the DIO driver works on pins and ports which are configured by the PORT driver. The PORT driver is initialized prior to the use of the DIO functions. Otherwise, DIO functions will exhibit undefined behavior.

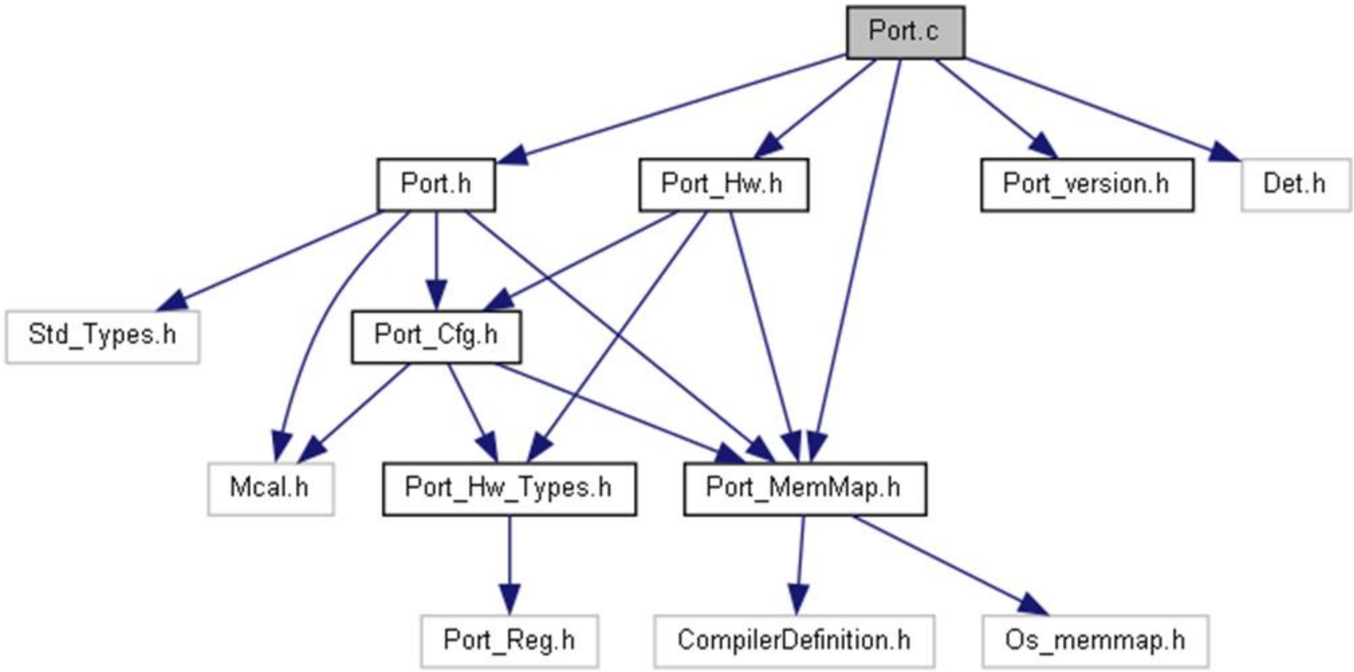
The hardware configured by the Port driver is PORT (Port Control).

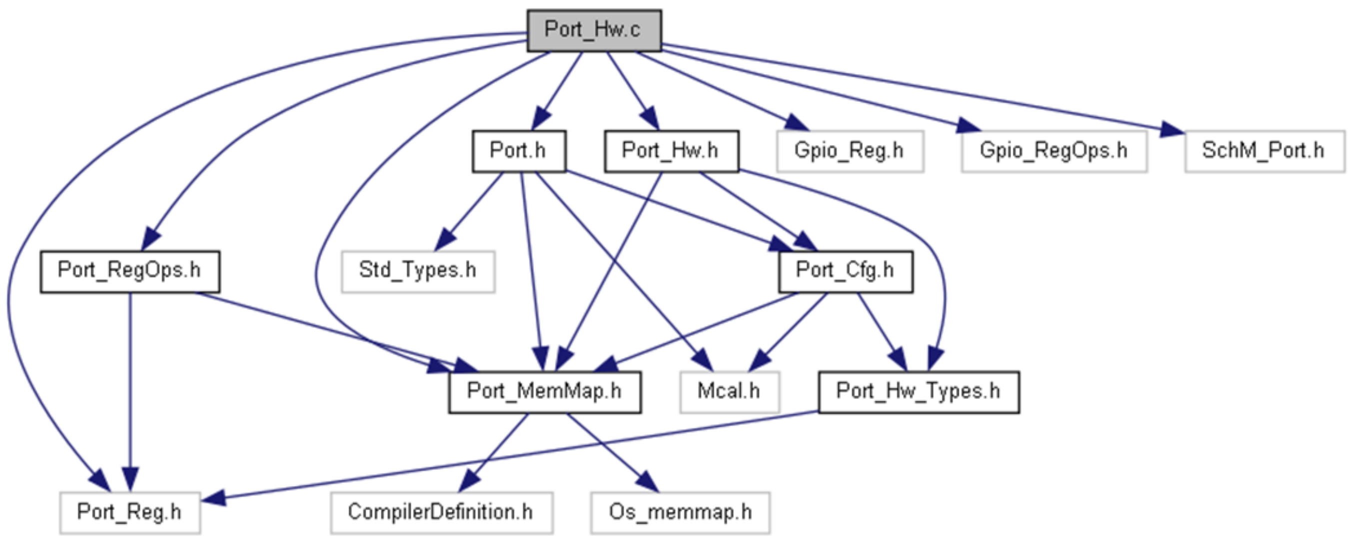
Chapter 2 Software Design

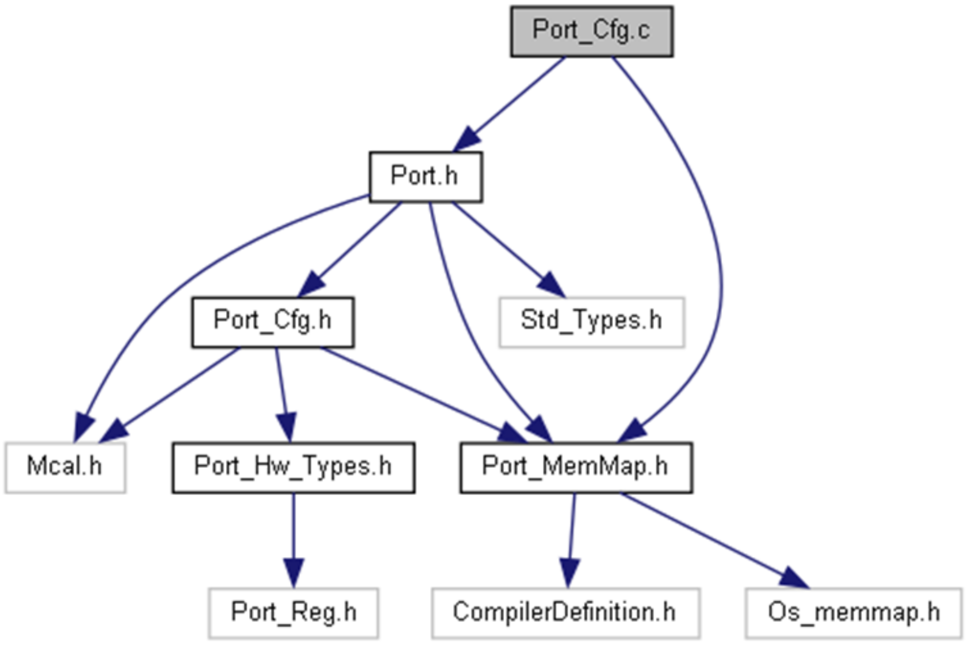
2.1 Rejected Requirements

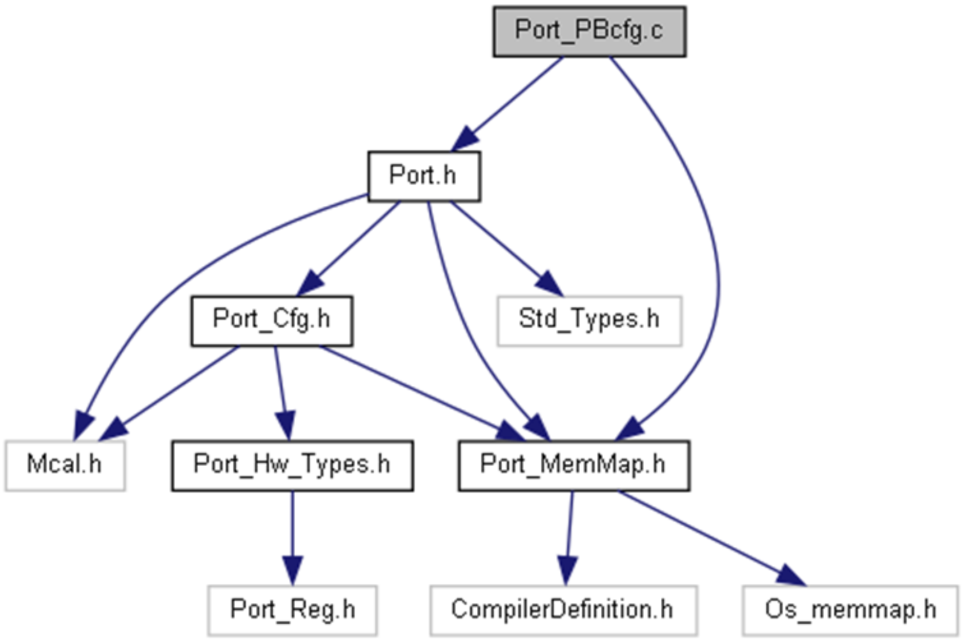
Rejected Requirement	
Description	Port_Lcfg.c shall include Port_MemMap.h and Port.h
Rejection Reason	Currently no support for link-time configuration is provided.

2.2 File Structure









2.3 Macros

2.3.1 Macros in Port.h

- #define **PORT_INSTANCE_ID** ((uint8)0x0)
  Instance ID of port drive.

- #define **PORT_INIT_ID** ((uint8)0x00)
  API service ID for PORT Init function.

- #define **PORT_SETPINDIRECTION_ID** ((uint8)0x01)
  API service ID for PORT set pin direction function.

- #define **PORT_REFRESHPINDIRECTION_ID** ((uint8)0x02)
  API service ID for PORT refresh pin direction function.

- #define **PORT_GETVERSIONINFO_ID** ((uint8)0x03)
  API service ID for PORT get version info function.

- #define **PORT_SETPINMODE_ID** ((uint8)0x04)
  API service ID for PORT set pin mode.

- #define **PORT_E_PARAM_PIN** ((uint8)0x0A)
  Invalid Port Pin ID requested.

- #define **PORT_E_DIRECTION_UNCHANGEABLE** ((uint8)0x0B)
  Port Pin Direction not configured as changeable.

- #define **PORT_E_INIT_FAILED** ((uint8)0x0C)
  API Port_Init( ) service called with wrong parameters

- #define **PORT_E_PARAM_INVALID_MODE** ((uint8)0x0D)
  API Port_SetPinMode() service called when the mode is invalid

- #define **PORT_E_MODE_UNCHANGEABLE** ((uint8)0x0E)
  API Port_SetPinMode() service called when the mode is unchangeable.

- #define **PORT_E_UNINIT** ((uint8)0x0F)
  API service called without module initialization.

- #define **PORT_E_PARAM_POINTER** ((uint8)0x10)
  API service called with NULL Pointer Parameter.

- #define **PORT_E_GET_SPIN_LOCK_FAILED** ((uint8)0x11)
  API service called with get spin lock failed.

- #define **PORT_SPIN_LOCK_TIMEOUT** ((uint16)0xFFFFU)
  Port spin lock timeout.

2.3.2 Macros in Port_version.h

- #define **PORT_AR_RELEASE_MAJOR_VERSION** 4

- #define **PORT _AR_RELEASE_MINOR_VERSION** 6

- #define **PORT _AR_RELEASE_REVISION_VERSION** 0

- #define **PORT_MODULE_ID** 124
- #define **PORT_SW_MAJOR_VERSION** 1
- #define **PORT_SW_MINOR_VERSION** 1
- #define **PORT_SW_PATCH_VERSION** 0
- #define **PORT_VENDOR_ID 174**

2.3.3 Macros in Port_Cfg.h

- #define **PORT_CFG_AR_RELEASE_MAJOR_VERSION** 4
- #define **PORT_CFG_AR_RELEASE_MINOR_VERSION** 6
- #define **PORT_CFG_AR_RELEASE_REVISION_VERSION** 0
- #define **PORT_CFG_SW_MAJOR_VERSION** 1
- #define **PORT_CFG_SW_MINOR_VERSION** 1
- #define **PORT_CFG_SW_PATCH_VERSION** 0
- #define **PORT _CFG_VENDOR_ID** 174
- #define **PORT_ALT0_FUNC_MODE** ((Port_PinModeType)0)
  Port Alternate 0 Mode.
- #define **PORT_GPIO_MODE** ((Port_PinModeType)1)
  Port GPIO Mode.
- #define **PORT_ALT2_FUNC_MODE** ((Port_PinModeType)2)
  Port Alternate 2 Mode.
- #define **PORT_ALT3_FUNC_MODE** ((Port_PinModeType)3)
  Port Alternate 3 Mode.
- #define **PORT_ALT4_FUNC_MODE** ((Port_PinModeType)4)
  Port Alternate 4 Mode.
- #define **PORT_ALT5_FUNC_MODE** ((Port_PinModeType)5)
  Port Alternate 5 Mode.

- #define **PORT_ALT6_FUNC_MODE** ((Port_PinModeType)6)
  Port Alternate 6 Mode.

- #define **PORT_ALT7_FUNC_MODE** ((Port_PinModeType)7)
  Port Alternate 7 Mode.

- #define **PORT_DEV_ERROR_DETECT** (STD_ON)
  Enable/disable development error detection.

- #define **PORT_SET_PIN_DIRECTION_API** (STD_ON)
  Use/remove Port_SetPinDirection function from the compiled driver.

- #define **PORT_SET_PIN_MODE_API** (STD_ON)
  Use/remove Port_SetPinMode function from the compiled driver.

- #define **PORT_FREEZE_JTAG_AND_RESET_PINS_API** STD_OFF
  Enable/disable the configuration of JTAG and reset Pins(PTA4, PTA5, PTA10, PTC4, PTC5).

- #define **PORT_ENABLE_USER_MODE_SUPPORT** (STD_OFF)

- #define **PORT_VERSION_INFO_API** (STD_ON)
  PORT_VERSION_INFO_API switch.

- #define **PortConfigSet_PortContainer_0_PortPin_0** (Port_PinType)31
  Port pin symbolic names.

- #define **PIN_MODE_OPTIONS_U8** ((uint8)8)
  Number of available pad modes options.

- #define **MAX_PORT_PIN_NUM_U16** ((uint16)154)
  The maximum number of port Pins supported by the platform.

- #define **MAX_CONFIGURED_PORTA_U8** ((uint8)25)
  The maximum number of configured Pins in PortA.

- #define **MAX_CONFIGURED_PORTB_U8** ((uint8)25)
  The maximum number of configured Pins in PortB.

- #define **MAX_CONFIGURED_PORTC_U8** ((uint8)25)
  The maximum number of configured Pins in PortC.

- #define **MAX_CONFIGURED_PORTD_U8** ((uint8)25)
  The maximum number of configured Pins in PortD.

- #define **MAX_CONFIGURED_PORTE_U8** ((uint8)24)
  The maximum number of configured Pins in PortE.

- #define **MAX_CONFIGURED_PINS_U16** ((uint16)124)
  The maximum channel number of configured pins.

- #define **MAX_CONFIGURED_DIGITAL_FILTER_PORTS_U8** (0U)
  The number of configured digital filter ports.

- #define **PORT_PRECOMPILE_SUPPORT** (STD_OFF)
  Port driver PRECOMPILE configuration switch.

- #define **PORT_SUPPORT_MULTICORE** (STD_ON)
  Port driver support multicore.

- #define **PORT_MAX_NUMBER** (9u)
  Port max number

2.4 Enums

2.4.1 Enumerations in Port_Hw_Types.h

2.4.1.1 Port_Number

Enumeration	Port_Number	
Description	This enumerated type allows the selection of PortA ~ PortE.	
Values	**Value**	**Description**
	Port_A = 0	PortA instance
	Port_B = 1	PortB instance
	Port_C = 2	PortC instance
	Port_D = 3	PortD instance
	Port_E = 4	PortE instance
	Port_F = 5	PortF instance
	Port_G = 6	PortG instance
	Port_H = 7	PortH instance
	Port_I= 8	PortI instance

2.4.1.2 Port_PinDirectionType

Enumeration	Port_PinDirectionType	
Description	This enumerated type allows the selection of different Pin directions.	
Values	**Value**	**Description**
	PORT_PIN_DISABLE = 0	No settings: the pin is not available.
	PORT_PIN_IN = 1	Sets port pin as input.
	PORT_PIN_OUT = 2	Sets port pin as output.
	PORT_PIN_HIGH_Z = 3	Sets port pin as high-Z.

2.4.1.3 Port_LockType

Enumeration	Port_LockType

Description	This enumerated type indicates the acquisition lock return status.	
Values	**Value**	**Description**
	GET_LOCK_FAIL = 0	Get lock fail.
	GET_LOCK_SUCCESS = 1	Get lock success.

2.5 Typedefs

2.5.1 Typedefs in Port_Hw_Types.h

- typedef uint8 **Port_PinNumber**
  Pin number type.

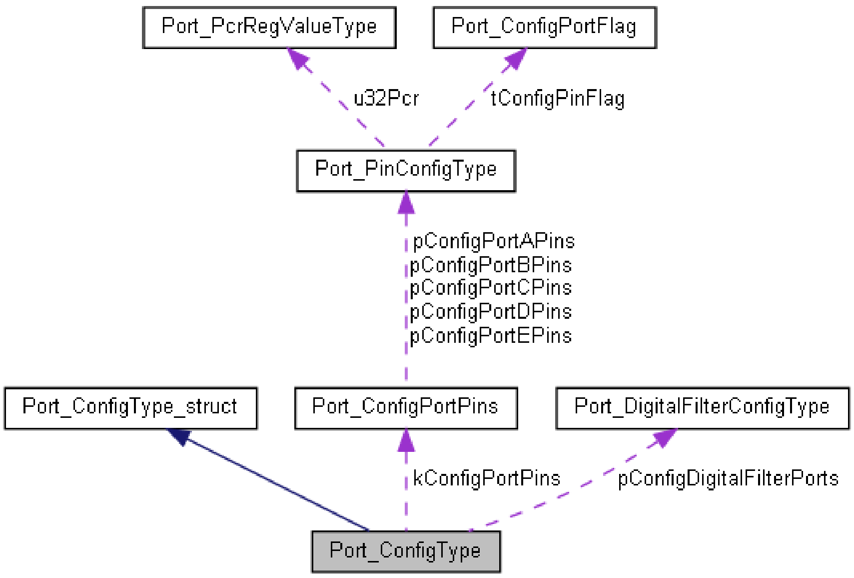
- typedef uint16 **Port_PinType**
  Data type for the symbolic name of a port pin.

- typedef uint8 **Port_PinModeType**
  Different port pin modes.

- typedef uint32 **Port_RegValueType**
  Port register value.

2.6 Structures

2.6.1 Port_ConfigType

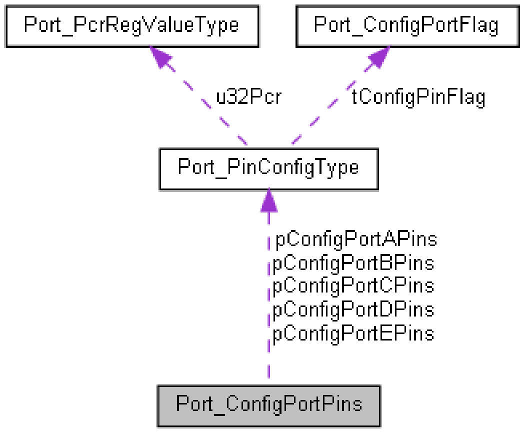
<table>
<tr><td>Structure</td><td>Port_ConfigType</td></tr>
<tr><td>Description</td><td>Type of the external data structure containing the initialization data for this module.</td></tr>
<tr><td>Diagram</td><td>
</td></tr>
<tr><td>Data Fields</td><td>• const Port_ConfigPortPins kConfigPortPins
Structure containing configuration information of the Port pins.
• const Port_DigitalFilterConfigType *pConfigDigitalFilterPorts
Pointer to the digital filter configuration of Port.</td></tr>
</table>

2.6.2 Port_DigitalFilterConfigType

Structure	Port_DigitalFilterConfigType
Description	This structure contains all configuration information of a digital filter port.

Diagram	N/A
Data Fields	• uint8 u8FPort Digital filter port. • uint8 u8FClock Select digital filter clock source. • uint8 u8FWidth Digital filter width. • uint32 u32FPinMask Mask of pins for which digital filter is enabled.

2.6.3 Port_ConfigPortPins

Structure	Port_ConfigPortPins
Description	Structure containing the configuration of all port pins.
Diagram	
Data Fields	• const Port_PinConfigType *pConfigPortAPins Pointer to PortA pins configuration. • const Port_PinConfigType *pConfigPortBPins Pointer to PortB pins configuration. • const Port_PinConfigType *pConfigPortCPins Pointer to PortC pins configuration. • const Port_PinConfigType *pConfigPortDPins Pointer to PortD pins configuration. • const Port_PinConfigType *pConfigPortEPins Pointer to PortE pins configuration.

2.6.4 Port_PinConfigType

Structure	Port_PinConfigType
Description	This structure contains all configuration parameters of a single pin.
Diagram	N/A
Data Fields	• Port_PinNumber u8Pin Pin defined on Port. • Port_PcrRegValueType u32Pcr PCR register. • uint8 u8Pdo Pad data output. • Port_PinDirectionType ePddr Pad data direction.

	• Port_ConfigPortFlag tConfigPinFlag Direction and mode changeability.

2.6.5 Port_ConfigPortFlag

Structure	Port_ConfigPortFlag
Description	This flag indicates some Port Pin status.
Diagram	N/A
Data Fields	• boolean bDirChangableFlag Direction changeability. • boolean bModeChangableFlag Mode changeability.

2.6.6 Port_Type

Structure	Port_Type
Description	This structure contains registers value of each Port.
Diagram	N/A
Data Fields	• __IO uint32 PCR[PORT_PCR_COUNT] Port Control Register, offset: 0x0. • __O uint32 GPCLR Global Pin Control Low Register, offset: 0x80. • __O uint32 GPCHR. Global Pin Control High Register, offset: 0x84. • __O uint32 GICLR Global Interrupt Control Low Register, offset: 0x88. • uint8 RESERVED_0[16] • __IO uint32 ISFR Interrupt Status Flag Register, offset: 0xA0. • __IO uint32 DFER Digital Filter Enable Register, offset: 0xC0. • __IO uint32 DFCR Digital Filter Clock Register, offset: 0xC4. • __IO uint32 DFWR Digital Filter Width Register, offset: 0xC8.

2.6.7 Gpio_Type

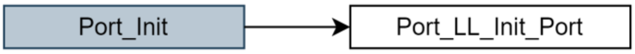
Structure	Gpio_Type
Description	Register layout typedef.
Diagram	N/A
Data Fields	• volatile uint32 PDOR Port Data Output Register, offset: 0x0. • volatile uint32 PSOR Port Set Output Register, offset: 0x4. • volatile uint32 PCOR Port Clear Output Register, offset: 0x8. • volatile uint32 PTOR

	Port Toggle Output Register, offset: 0xc. • volatile uint32 PDIR Port Data Input Register, offset: 0x10. • volatile uint32 PDDR Port Data Direction Register, offset: 0x14. • volatile uint32 PIDR Port Input Disable Register, offset: 0x18.

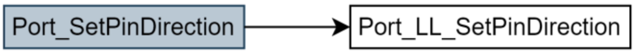
2.7 API Functions

2.7.1 Functions in Port.h

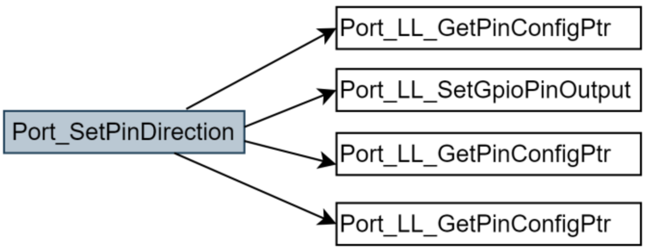
2.7.1.1 void Port_Init(const Port_ConfigType *ConfigPointer)

Function	void Port_Init(const Port_ConfigType *ConfigPointer)	
Description	Initializes the Port drive module.	
Diagram		
Parameters	**Parameter**	**Description**
	ConfigPointer	Pointer to configuration set.
Returns	N/A	

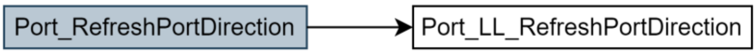
2.7.1.2 void Port_SetPinDirection(Port_PinType Pin, Port_PinDirectionType Direction)

Function	void Port_SetPinDirection(Port_PinType Pin, Port_PinDirectionType Direction)	
Description	Sets the port pin direction.	
Diagram		
Parameters	**Parameter**	**Description**
	Pin	Port Pin ID number.
	Direction	Port Pin direction.
Returns	N/A	

2.7.1.3 void Port_SetPinMode(Port_PinType PinId, Port_PinModeType PinMuxMode)

Function	void Port_SetPinMode(Port_PinType PinId, Port_PinModeType PinMuxMode)	
Description	Sets the port pin mode.	
Diagram		
Parameters	**Parameter**	**Description**
	PinId	Port Pin ID number.
	PinMuxMode	New Port Pin mode to be set on port pin
Returns	N/A	

2.7.1.4 void Port_RefreshPortDirection(void)

Function	void Port_RefreshPortDirection(void)	
Description	Refreshes port direction.	
Diagram		
Parameters	N/A	
Returns	N/A	

2.7.1.5 void Port_GetVersionInfo(Std_VersionInfoType *versioninfo)

Function	void Port_GetVersionInfo(Std_VersionInfoType *versioninfo)	
Description	Returns the version information of this module.	
Diagram	N/A	
Parameters	**Parameter**	**Description**
	versioninfo	Version information structure pointer.
Returns	N/A	

2.8 Hardware Functions

2.8.1 Functions in Port_Hw.c

2.8.1.1 Port_PinConfigType *Port_LL_GetPinConfigPtr(Port_Number ePort, Port_PinNumber u8Pin, const Port_ConfigType *pConfigPorts)

Function	Port_PinConfigType *Port_LL_GetPinConfigPtr(Port_Number ePort, Port_PinNumber u8Pin, const Port_ConfigType *pConfigPorts)	
Description	Initializes pins of one Port.	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	u8Pin	Pin number defied on Port.
	pConfigPorts	Pointer containing configuration information of one pin.
Returns	N/A	
Referenced By	Port_SetPinDirection(), Port_SetPinMode()	

2.8.1.2 void Port_LL_Init_Port(Port_Number u8Port, const Port_PinConfigType *pConfigPorts, uint8 u8PinNum)

Function	void Port_LL_Init_Port(Port_Number u8Port, const Port_PinConfigType *pConfigPorts, uint8 u8PinNum)	
Description	Initializes pins of one Port.	
Parameters	**Parameter**	**Description**
	u8Port	Port number.
	pConfigPorts	Ports configuration pointer.
	u8PinNum	The number of pins that need to be initialized in a Port.
Returns	N/A	
Referenced By	Port_Init ()	

2.8.1.3 void Port_LL_SetPinDirection(Port_Number ePort, Port_PinNumber u8Pin, Port_PinDirectionType eDirection)

Function	void Port_LL_SetPinDirection(Port_Number ePort, Port_PinNumber u8Pin, Port_PinDirectionType eDirection)

Description	Sets Pin direction.	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	u8Pin	Pin number defined on Port.
	eDirection	Pin direction.
Returns	N/A	
Referenced By	Port_SetPinDirection()	

2.8.1.4 void Port_LL_SetPinMode(Port_Number ePort, Port_PinNumber u8PinMode, const Port_PinConfigType *pConfigOnePin)

Function	void Port_LL_SetPinMode(Port_Number ePort, Port_PinModeType u8PinMode, const Port_PinConfigType *pConfigOnePin)	
Description	Sets Mode for a Pin.	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	u8PinMode	Pin mode.
	pConfigOnePin	Pointer containing configuration information of one pin.
Returns	N/A	
Referenced By	Port_SetPinMode()	

2.8.1.5 void Port_LL_RefreshPortDirection(Port_Number ePort, const Port_PinConfigType *pPortPinsConfigInfo, uint8 u8PortPinsCount)

Function	void Port_LL_RefreshPortDirection(Port_Number ePort, const Port_PinConfigType *pPortPinsConfigInfo,uint8 u8PortPinsCount)	
Description	Refreshes Port direction.	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	pPortPinsConfigInfo,	Pointer containing information Ports configuration.
	u8PortPinsCount	Pins count in this Port.
Returns	N/A	
Referenced By	Port_RefreshPortDirection()	

2.8.1.6 void Port_LL_SetGpioPinOutput(Port_Number ePort, Port_PinNumber u8Pin, const Port_PinConfigType *pConfigOnePin)

Function	void Port_LL_SetGpioPinOutput(Port_Number ePort, Port_PinNumber u8Pin, const Port_PinConfigType *pConfigOnePin)	
Description	Sets the Pin output level.	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	u8Pin	Pin number defined on Port.
	pConfigOnePin	Pointer containing configuration information of one pin.
Returns	N/A	
Referenced By	Port_SetPinMode()	

2.8.1.7 Port_LockType Port_LL_GetLock(Port_Number ePort, Port_PinNumber u8Pin)

Function	Port_LockType Port_LL_GetLock(Port_Number ePort, Port_PinNumber u8Pin)

Description	Get Lock	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	u8Pin	Pin ID.
Returns	Port_LockType	
Referenced By	Port_GetLock()	

2.8.1.8 void Port_LL_ReleaseLock(Port_Number ePort, const Port_RegValueType u32UsedCoreId, Port_PinNumber u8Pin)

Function	void Port_LL_ReleaseLock(Port_Number ePort, const Port_PinConfigType *pConfigOnePin, Port_PinNumber u8Pin)	
Description	Release Lock	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	u32UsedCoreId	Core with permissions.
	u8Pin	Pin ID.
Returns	N/A	
Referenced By	Port_ReleaseLock()	

2.8.1.9 Std_ReturnType Port_GetLock(uint8 u8ServiceID, Port_Number ePort, Port_PinNumber u8Pin)

Function	Std_ReturnType Port_GetLock(uint8 u8ServiceID, Port_Number ePort, Port_PinNumber u8Pin)	
Description	Get the lock of the Port	
Parameters	**Parameter**	**Description**
	u8ServiceID	The service id of the caller function.
	ePort	Port number.
	u8Pin	Pin ID.
Returns	Std_ReturnType	
Referenced By	Port_SetPinDirection(), Port_SetPinMode(), Port_LL_RefreshPortDirection()	

2.8.1.10 void Port_ReleaseLock(Port_Number ePort, const Port_RegValueType u32UsedCoreId, Port_PinNumber u8Pin)

Function	void Port_ReleaseLock(Port_Number ePort, const Port_RegValueType u32UsedCoreId, Port_PinNumber u8Pin)	
Description	Release the Port lock	
Parameters	**Parameter**	**Description**
	ePort	Port number.
	u32UsedCoreId	Core with permissions.
	u8Pin	Pin ID.
Returns	Port_LockType	
Referenced By	Port_SetPinDirection(), Port_SetPinMode(), Port_LL_RefreshPortDirection()	

2.9 Peripheral Functions

2.9.1 Functions in Port_RegOps.h

2.9.1.1 LOCAL_INLINE void PORT_HWA_ConfigPin(PORT_Type *pPort, uint8 u8Pin, uint32 u32PcrReg)

Function	LOCAL_INLINE void PORT_HWA_ConfigPin(PORT_Type *pPort, uint8 u8Pin, uint32 u32PcrReg)	
Description	Configures the Pin PCR value.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	u8Pin	Pin number defined on Port. (0~31)
	u32PcrReg	PCR register value of the Pin.
Returns	N/A	
Referenced By	Port_LL_Init_Port(), Port_LL_SetPinMode()	

2.9.1.2 LOCAL_INLINE uint32 PORT_HWA_ReadPcrValue(PORT_Type *pPort, uint8 u8Pin)

Function	LOCAL_INLINE uint32 Port_HWA_ReadPcrValue(PORT_Type *pPort, uint8 u8Pin)	
Description	Gets the Port PCR register value.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Port_LL_SetPinMode()	

2.9.1.3 LOCAL_INLINE void PORT_HWA_SetPinPullEnable(PORT_Type *pPort, uint8 u8Pin, uint8 u8PeValue)

Function	LOCAL_INLINE void PORT_HWA_SetPinPullEnable(PORT_Type *pPort, uint8 u8Pin, uint8 u8PeValue)	
Description	Sets pin pull enable.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	u8Pin	Pin number defined on Port. (0~31)
	u8PeValue	Pin pull enable configuration value. (1/0)
Returns	N/A	
Referenced By	Port_LL_SetPinMode()	

2.9.1.4 LOCAL_INLINE void PORT_HWA_SetPinPullMode(PORT_Type *pPort, uint8 u8Pin, uint8 u8PsValue)

Function	LOCAL_INLINE void PORT_HWA_SetPinPullMode(PORT_Type *pPort, uint8 u8Pin, uint8 u8PsValue)	
Description	Sets pin pull mode.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	u8Pin	Pin number defined on Port. (0~31)
	u8PsValue	Pin pull select configuration value. 0 – Enable internal pulldown resistor. 1 - Enable internal pullup resistor.
Returns	N/A	
Referenced By	Port_LL_SetPinMode()	

2.9.1.5 LOCAL_INLINE void PORT_HWA_SetDigitalFilterClkSrc(PORT_Type *pPort, uint8 eClkSrc)

Function	LOCAL_INLINE void PORT_HWA_SetDigitalFilterClkSrc(PORT_Type *pPort, uint8 eClkSrc)	
Description	Sets the digital filter clock source.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	eClkSrc	Clock source. 0 - Select the bus clock as the clock source. 1 - Select the AON32K_CLK clock as the clock source.
Returns	N/A	
Referenced By	Port_HL_Init()	

2.9.1.6 LOCAL_INLINE void PORT_HWA_ConfigDigitalFilterWidth(PORT_Type *pPort, uint32 u32FilterWidth)

Function	LOCAL_INLINE void PORT_HWA_ConfigDigitalFilterWidth(PORT_Type *pPort, uint32 u32FilterWidth)	
Description	Configures the digital filter width of Port.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	u32FilterWidth	Digital filter width (value range:0-31).
Returns	N/A	
Referenced By	Port_HL_Init()	

2.9.1.7 LOCAL_INLINE void PORT_HWA_SetDigitalFilterEnable(PORT_Type *pPort, uint32 u32FilterPinMask)

Function	LOCAL_INLINE void PORT_HWA_SetDigitalFilterEnable(PORT_Type *pPort, uint32 u32FilterPinMask)	
Description	Sets digital filter enable.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	u32FilterPinMask	Digital filter pin mask.
Returns	N/A	
Referenced By	Port_HL_Init()	

2.9.1.8 LOCAL_INLINE void PORT_HWA_SetDWP (PORT_Type *pPort, uint8 u8Pin, uint8 u8PsValue)

Function	LOCAL_INLINE void PORT_HWA_SetDWP(PORT_Type *pPort, uint8 u8Pin, uint8 u8PsValue)	
Description	Sets pin direction.	
Parameters	**Parameter**	**Description**
	pPort	Port instance.
	u8Pin	Pin number defined on Port. (0~31)
	u8PsValue	Pin DWP configuration value.
Returns	N/A	
Referenced By	Port_LL_GetLock (),Port_LL_ReleaseLock()	

2.9.2 Functions in GPIO_RegOps.h

2.9.2.1 LOCAL_INLINE void GPIO_HWA_SetPinOutput(GPIO_Type *pGpio, uint8 u8Pin)

Function	LOCAL_INLINE void GPIO_HWA_SetPinOutput(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Sets the pin output.	
Parameters	**Parameter**	**Description**
	pGpio	GPIO instance.

	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Port_LL_Init_Port()	

2.9.2.2 LOCAL_INLINE void GPIO_HWA_ClearPinOutput(GPIO_Type *pGpio, uint8 u8Pin)

Function	LOCAL_INLINE void GPIO_HWA_ClearPinOutput(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Clears the pin output.	
Parameters	**Parameter**	**Description**
	pGpio	GPIO instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Port_LL_Init_Port(),Port_LL_SetGpioPinOutput()	

2.9.2.3 LOCAL_INLINE void GPIO_HWA_ClearPinDirection(GPIO_Type *pGpio, uint8 u8Pin)

Function	LOCAL_INLINE void GPIO_HWA_ClearPinDirection(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Clears pin direction.	
Parameters	**Parameter**	**Description**
	pGpio	GPIO instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Port_LL_Init_Port(),Port_LL_SetPinDirection()	

2.9.2.4 LOCAL_INLINE void GPIO_HWA_SetPinInputDisable(GPIO_Type *pGpio, uint8 u8Pin)

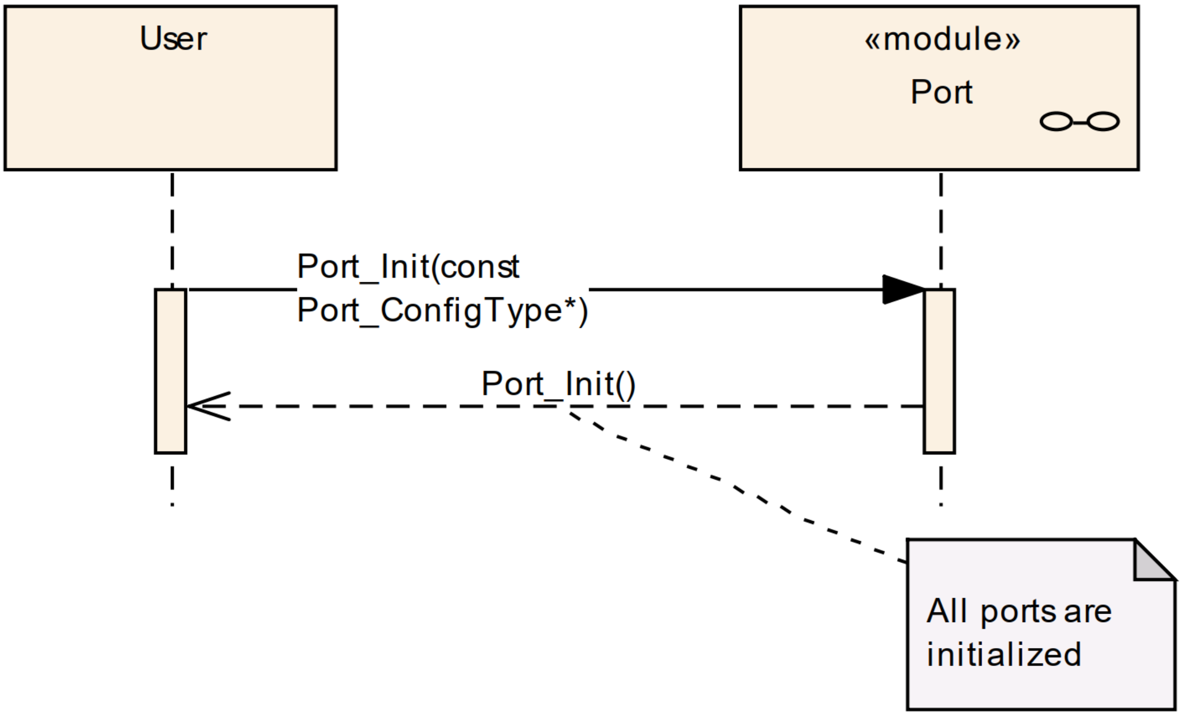
Function	LOCAL_INLINE void GPIO_HWA_SetPinInputDisable(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Disables the pin input.	
Parameters	**Parameter**	**Description**
	pGpio	GPIO instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Port_LL_Init_Port(),Port_LL_SetPinDirection(), Port_LL_RefreshPortDirection()	

2.9.2.5 LOCAL_INLINE void GPIO_HWA_SetPinDirection(GPIO_Type *pGpio, uint8 u8Pin)

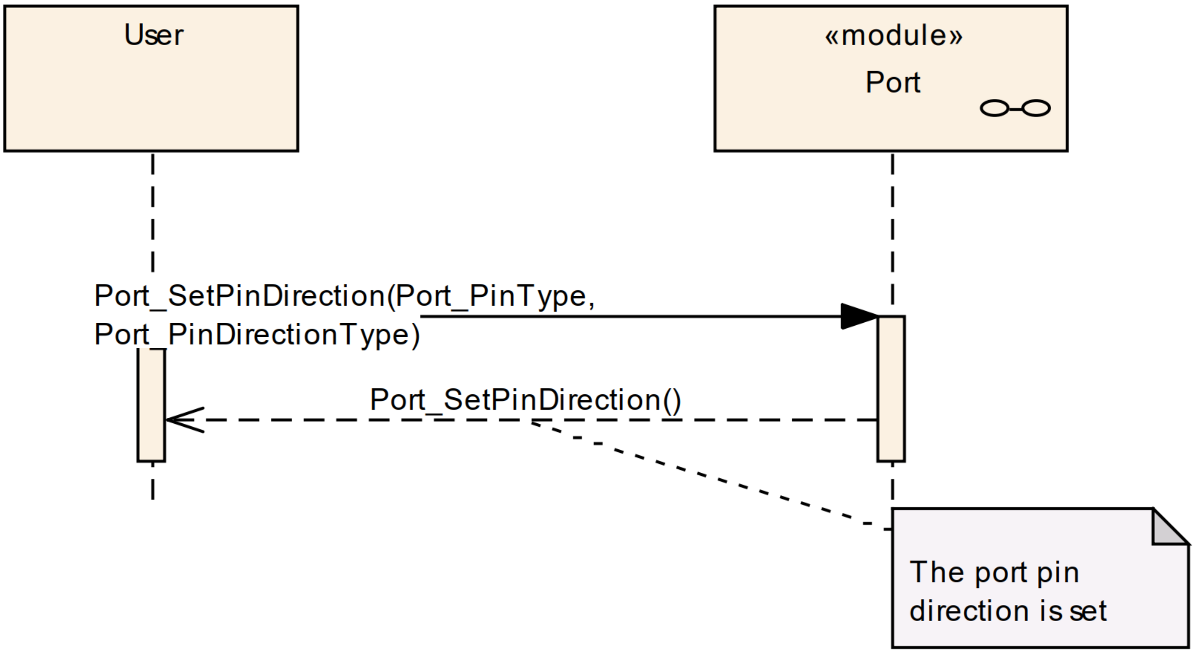
Function	LOCAL_INLINE void GPIO_HWA_SetPinDirection(GPIO_Type *pGpio, uint8 u8Pin)	
Description	Sets pin direction.	
Parameters	**Parameter**	**Description**
	pGpio	GPIO instance.
	u8Pin	Pin number defined on Port. (0~31)
Returns	N/A	
Referenced By	Port_LL_Init_Port(),Port_LL_SetPinDirection(),Port_LL_RefreshPortDirection()	

2.10 API Sequence Diagram

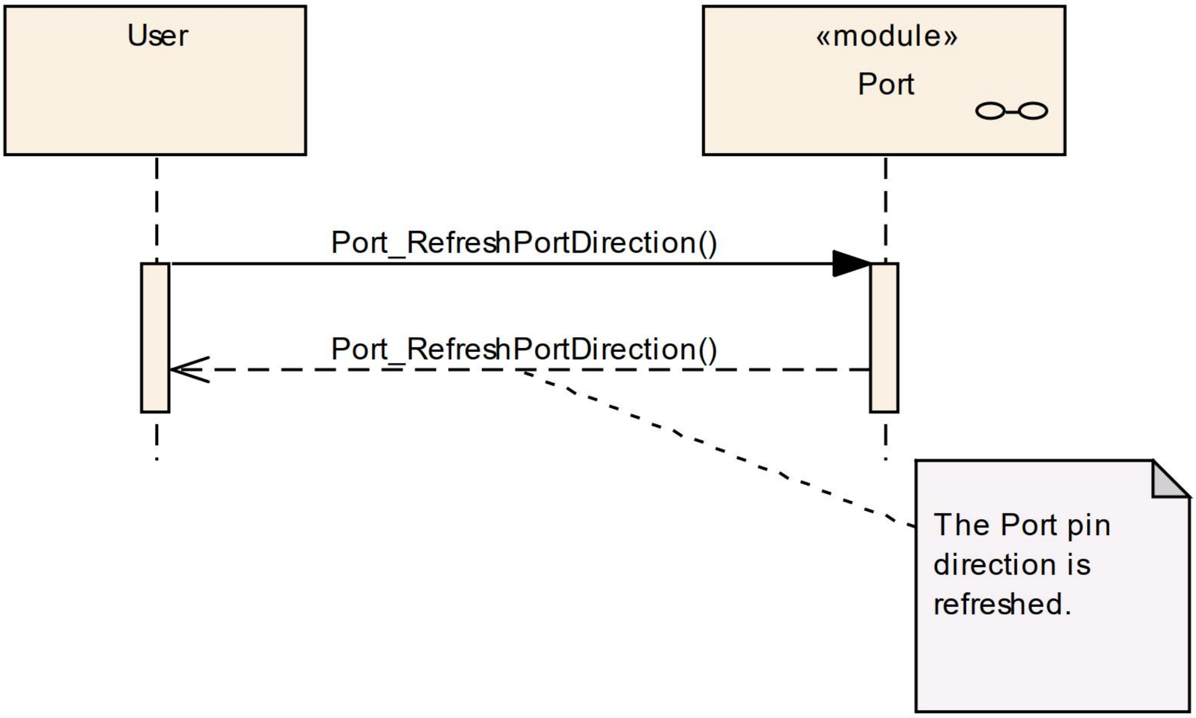
2.10.1 Overall Configuration of Ports



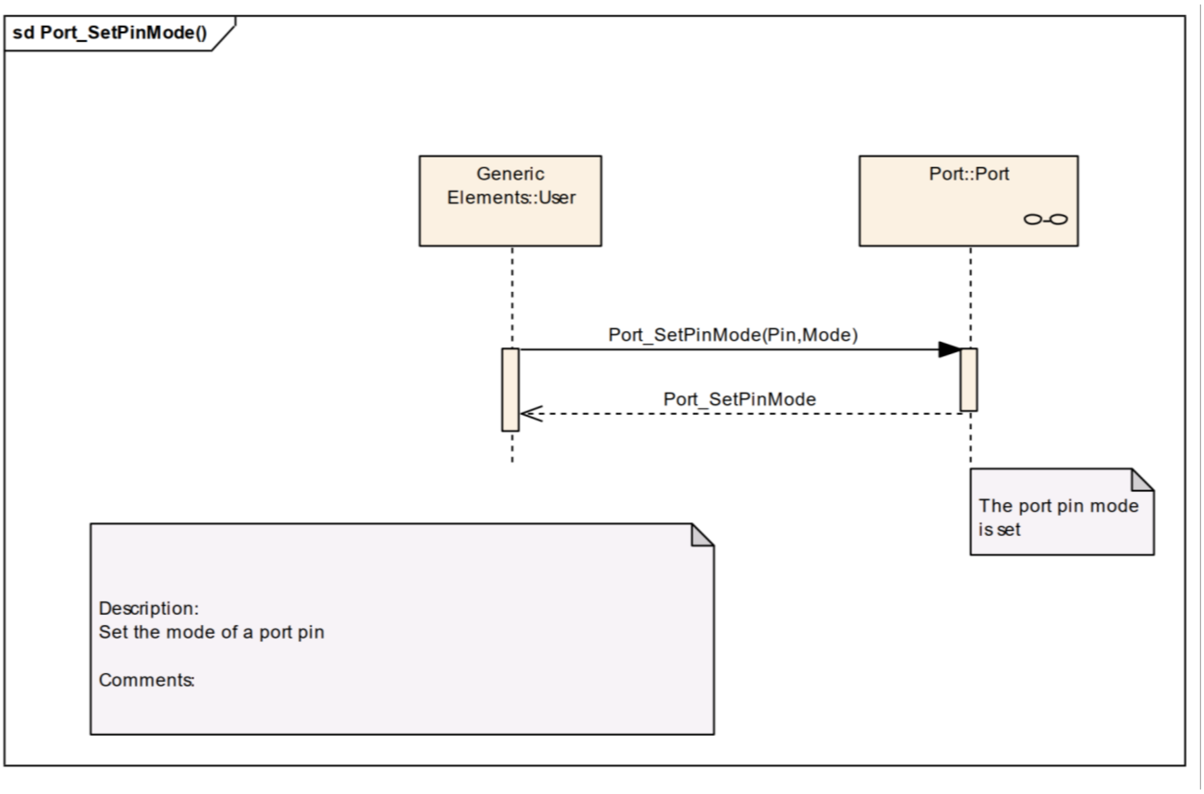
2.10.2 Set Direction of a Port Pin



2.10.3 Refresh Direction of All Port Pins



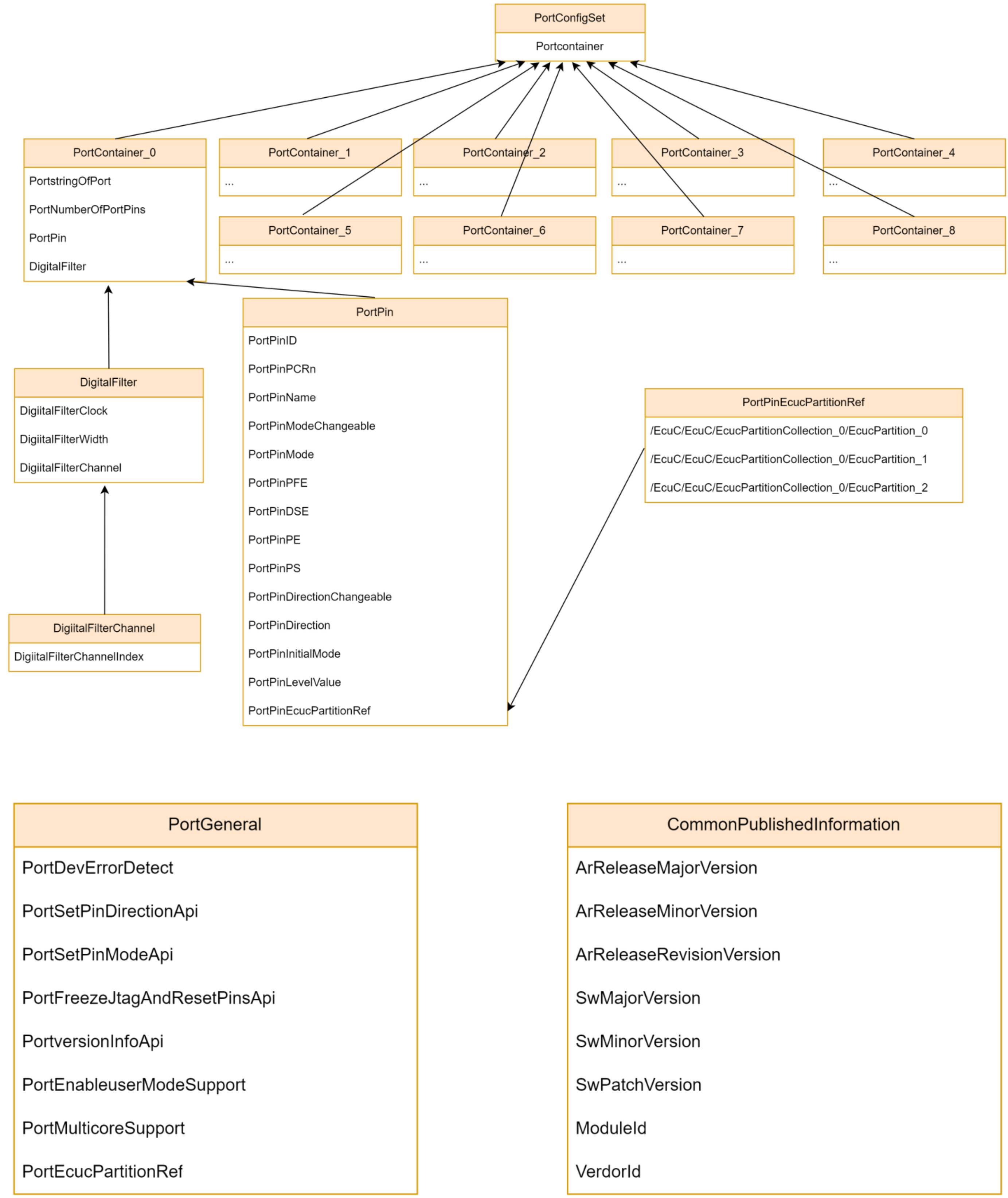
2.10.4 Change Mode of a Port Pin



Chapter 3 Tresos Configuration Items

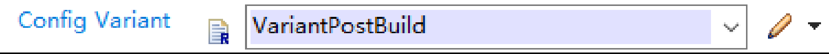
3.1 Container Inclusion Relation

The container inclusion relation is shown as below:

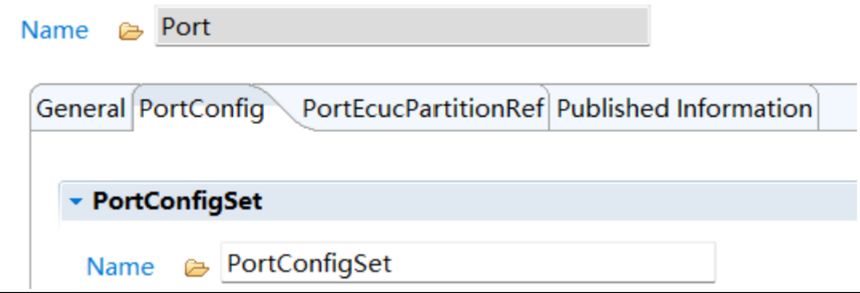


3.2 Containers and Variables

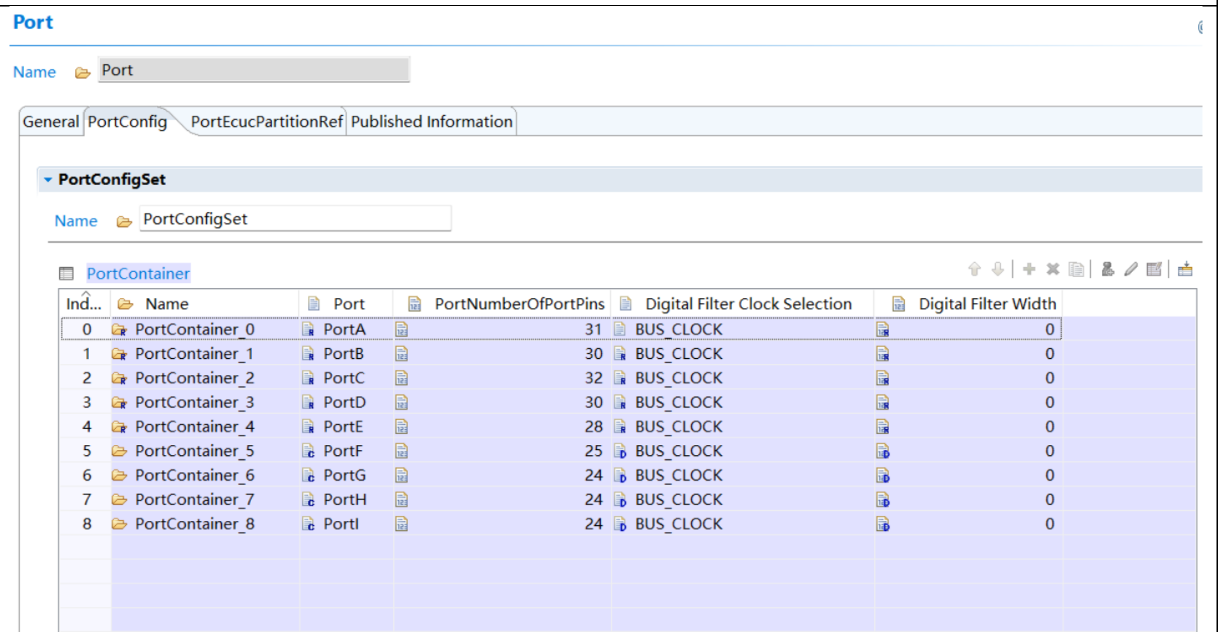
3.2.1 IMPLEMENTATION_CONFIG_VARIANT

<table>
<tr><td>Variables</td><td colspan="2">IMPLEMENTATION_CONFIG_VARIANT</td></tr>
<tr><td>Description</td><td colspan="2">N/A</td></tr>
<tr><td>Screenshot</td><td colspan="2">
</td></tr>
<tr><td rowspan="5">Properties</td><td>Property</td><td>Value</td></tr>
<tr><td>Type</td><td>Variable: enumeration</td></tr>
<tr><td>Label</td><td>Config Variant</td></tr>
<tr><td>Default</td><td>VariantPostBuild</td></tr>
<tr><td>Range</td><td>VariantPostBuild
VariantPreCompile</td></tr>
<tr><td>Returns</td><td colspan="2">N/A</td></tr>
</table>

3.2.2 PortConfigSet

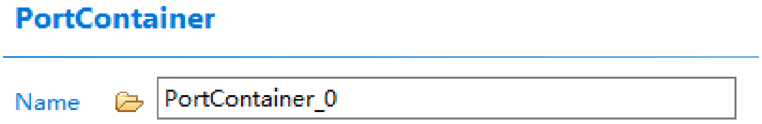
<table>
<tr><td>Container</td><td colspan="2">PortConfigSet</td></tr>
<tr><td>Description</td><td colspan="2">This container contains the configuration parameters and sub containers of the AUTOSAR Port module.</td></tr>
<tr><td>Screenshot</td><td colspan="2">
</td></tr>
<tr><td rowspan="2">Properties</td><td>Property</td><td>Value</td></tr>
<tr><td>Type</td><td>Container</td></tr>
</table>

3.2.2.1 PortContainer

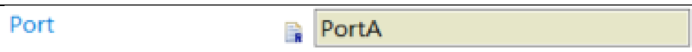
<table>
<tr><td>List</td><td colspan="2">PortContainer</td></tr>
<tr><td>Description</td><td colspan="2">N/A</td></tr>
<tr><td>Screenshot</td><td colspan="2">
</td></tr>
<tr><td>Properties</td><td>Property</td><td>Value</td></tr>
</table>

	Type	Map
	Min	5
	Max	5
	Tab	PortConfig

3.2.2.2 PortContainer

Container	PortContainer	
Description	Container collecting the PortPins.	
Screenshot		
Properties	**Property**	**Value**
	Type	Container

3.2.2.2.1 PortStringOfPort

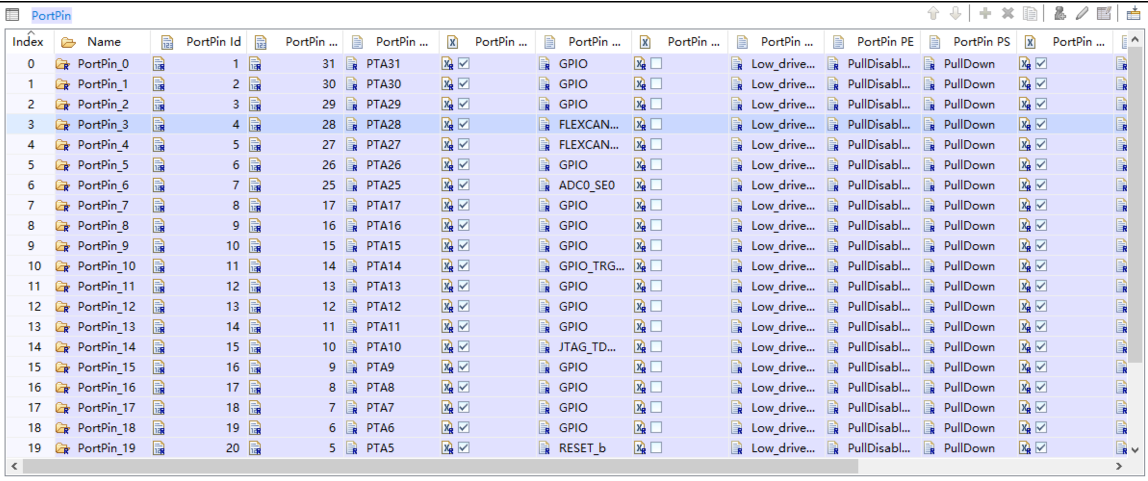
Variable	PortStringOfPort	
Description	The string indicates the Port name.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable:string
	Label	Port
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Editable	False

3.2.2.2.2 PortNumberOfPortPins

Variable	PortNumberOfPortPins	
Description	The number of specified PortPins in this PortContainer.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer
	Label	PortNumberOfPortPins
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Read Only	false
	Default	25

3.2.2.2.3 PortPin

List	PortPin
Description	The number of specified Port Pins in this Port Container.

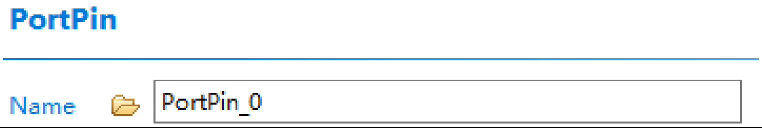
Screenshot	

PortPin

Index	Name	PortPin Id	PortPin ...	PortPin ...	PortPin ...	PortPin ...	PortPin ...	PortPin ...	PortPin PE	PortPin PS	PortPin ...
0	PortPin_0	1	31	PTA31	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
1	PortPin_1	2	30	PTA30	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
2	PortPin_2	3	29	PTA29	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
3	PortPin_3	4	28	PTA28	☑	FLEXCAN...	☐	Low_drive...	PullDisabl...	PullDown	☑
4	PortPin_4	5	27	PTA27	☑	FLEXCAN...	☐	Low_drive...	PullDisabl...	PullDown	☑
5	PortPin_5	6	26	PTA26	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
6	PortPin_6	7	25	PTA25	☑	ADC0_SE0	☐	Low_drive...	PullDisabl...	PullDown	☑
7	PortPin_7	8	17	PTA17	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
8	PortPin_8	9	16	PTA16	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
9	PortPin_9	10	15	PTA15	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
10	PortPin_10	11	14	PTA14	☑	GPIO_TRG...	☐	Low_drive...	PullDisabl...	PullDown	☑
11	PortPin_11	12	13	PTA13	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
12	PortPin_12	13	12	PTA12	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
13	PortPin_13	14	11	PTA11	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
14	PortPin_14	15	10	PTA10	☑	JTAG_TD...	☐	Low_drive...	PullDisabl...	PullDown	☑
15	PortPin_15	16	9	PTA9	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
16	PortPin_16	17	8	PTA8	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
17	PortPin_17	18	7	PTA7	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
18	PortPin_18	19	6	PTA6	☑	GPIO	☐	Low_drive...	PullDisabl...	PullDown	☑
19	PortPin_19	20	5	PTA5	☑	RESET_b	☐	Low_drive...	PullDisabl...	PullDown	☑

Properties	**Property**	**Value**
	Type	Map
	Min	Pin number of the corresponding Port.
	Max	Pin number of the corresponding Port.

3.2.2.2.4 PortPin

Container	PortPin
Description	Configuration of the individual port pins.
Screenshot	

Properties	**Property**	**Value**
	Type	Container

3.2.2.2.5 PortPinId

Variable	PortPinId
Description	Pin ID of the port pin. This value will be assigned to the symbolic name derived from the port pin container short name.
Screenshot	

Properties	**Property**	**Value**
	Type	Variable: Integer
	Label	PortPin ID
	Editable	false

3.2.2.2.6 PortPinPCRn

Variable	PortPinPCRn
Description	Used to specify the PCR (Port Control Register) for the configured pin.
Screenshot	

Properties	**Property**	**Value**
	Type	Variable: Integer
	Label	PortPin PCRn
	Origin	FLAGCHIP
	SymbolicNameValue	False

	Editable	False

3.2.2.2.7 PortPinName

Variable	PortPinName	
Description	Used to specify the port pin name.	
Screenshot	PortPin Name	PTA31
Properties	**Property**	**Value**
	Type	Variable:String
	Label	PortPin Name
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Editable	false

3.2.2.2.8 PortPinModeChangeable

Variable	PortPinModeChangeable	
Description	Parameter to indicate if the mode of a port pin is changeable during runtime. True: Port Pin mode changeable allowed. False: Port Pin mode changeable not permitted.	
Screenshot	PortPin Mode Changeable*	
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	PortPin Mode Changeable
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	False
	Default	true

3.2.2.2.9 PortPinMode

Variable	PortPinMode	
Description	Selects the PORT pin mode from the modes list. One or more modes may be valid for a pin.	
Screenshot	PortPin Mode	GPIO
Properties	**Property**	**Value**
	Type	Variable: Enumeration
	Label	PortPin Mode
	Origin	AUTOSAR_ECUC
	Default	GPIO

3.2.2.2.10 PortPinPFE

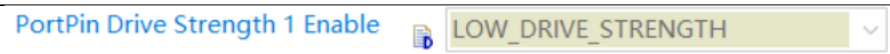
Variable	PortPinPFE	
Description	Passive Filter Enable	
Screenshot	PortPin Passive Filter Enable	
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	PortPin DSE
	Origin	FLAGCHIP
	SymbolicNameValue	false

	Default	false
	Editable	Be true only when: 1) The Pin have mode: "RESET_b" or "NMI_b" 2) Mode "RESET_b" or "NMI_b" is selected for the Pin.

3.2.2.2.11 PortPinDSE0

Variable	PortPinDSE0	
Description	Drive strength is valid if the pin is configured as a digital output(e.g. SPI_SOUT/UART_TX/FLEXCAN_TX...)	
Screenshot		
Properties	**Property**	**Value**
	Type	Enumeration
	Label	PortPin Drive Strength 0 Enable
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	Low_drive_strength
	Range	Low_drive_strength High_drive_strength

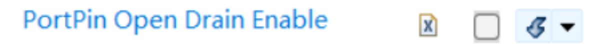
3.2.2.2.12 PortPinDSE1

Variable	PortPinDSE1	
Description	Drive strength is valid if the pin is configured as a digital output(e.g. SPI_SOUT/UART_TX/FLEXCAN_TX...)	
Screenshot		
Properties	**Property**	**Value**
	Type	Enumeration
	Label	PortPin Drive Strength 1 Enable
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	Low_drive_strength
	Range	Low_drive_strength High_drive_strength

3.2.2.2.13 PortPinSRE

Variable	PortPinSRE	
Description	Slew Rate is valid if the pin is configured as a digital function.	
Screenshot		
Properties	**Property**	**Value**
	Type	Enumeration
	Label	PortPin Slew Rate Enable
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	FASTEST_SLEW_RATE
	Range	FASTEST_SLEW_RATE SLOWEST_SLEW_RATE

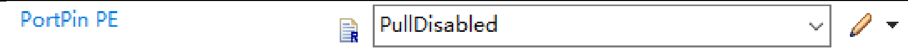
3.2.2.2.14 PortPinODE

Variable	PortPinODE	
Description	Open drain is valid if the pin is configured as a digital function.	
Screenshot		
Properties	**Property**	**Value**
	Type	Boolean
	Label	PortPin Open Drain Enable
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

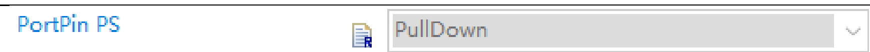
3.2.2.2.15 PortPinESTOP

Variable	PortPinESTOP	
Description	Emergency Stop is valid if the pin is configured as a digital function. If the FCSMU generates a fault and pin emergency stop is enabled, pin output function will be disabled.	
Screenshot		
Properties	**Property**	**Value**
	Type	Boolean
	Label	PortPin Emergency Stop Enable
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	False

3.2.2.2.16 PortPinPE

Variable	PortPinPE	
Description	Selects if the pull-up or pull-down resistors are enabled.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Enumeration
	Label	PortPin PE
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	PullDisabled
	Range	PullDisabled PullEnabled

3.2.2.2.17 PortPin PS

Variable	PortPin PS	
Description	Selects between the pull-up and pull-down resistors. Only valid when PortPin PE is set to "PullEnabled".	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Enumeration
	Label	PortPin PS
	Origin	FLAGCHIP
	SymbolicNameValue	false

	Editable	Only be true when PortPin PE is set to "PullEnabled".
	Default	PullDown
	Range	PullDown PullUp

3.2.2.2.18 PortPinDirectionChangeable

Variable	PortPinDirectionChangeable	
Description	Enable/Disable the changeability for the configured Pin. The checked box means the Direction Changeability is enabled. This is an implementation-specific parameter. The changeable pin direction can be set only for the GPIO pins. For a mode different than GPIO, pin direction changeability shall be disabled.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	PortPin Direction Changeable
	Origin	AUTOSAR_ECUC
	Default	true
	SymblicNameValue	false
	Editable	Only be true when Pin mode is set to GPIO.

3.2.2.2.19 PortPinDirection

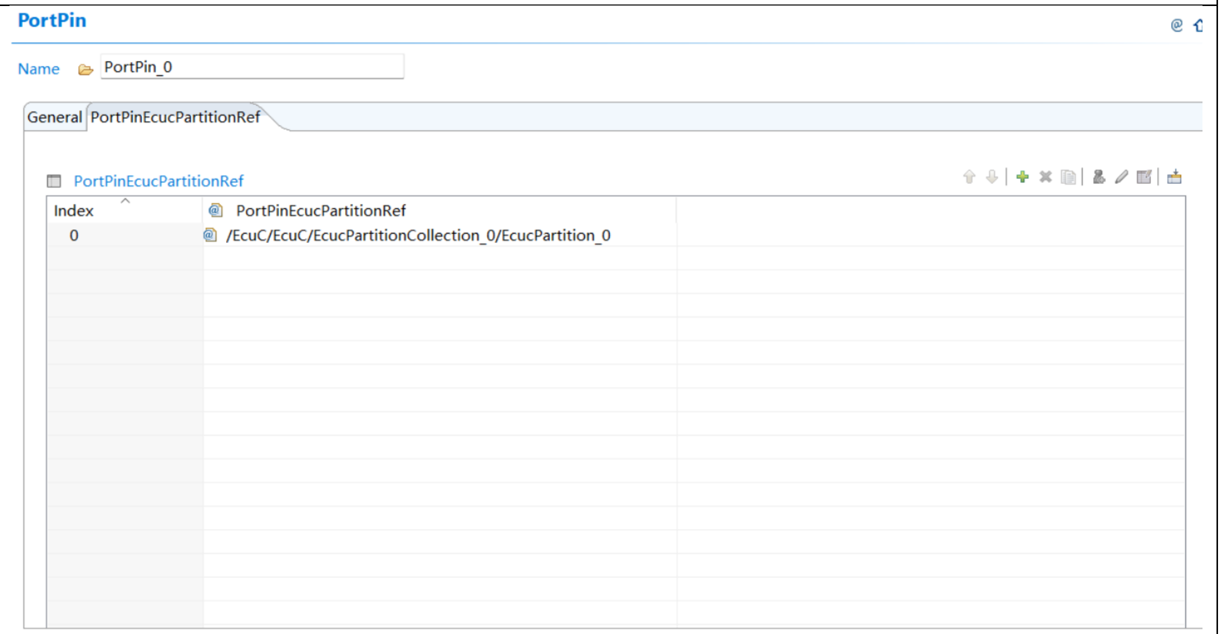
Variable	PortPinDirection	
Description	Selects the direction of the pin (IN, OUT, HIGH_Z) that will be configured by Port_Init() function if the pin is configured as GPIO. If the direction is not changeable, the value configured here is fixed. For the Alternative Function modes (PortPinMode is different from GPIO), the setting in this enumeration control is kept in the port configuration structure and it is used when Port_SetPinMode() is called at runtime to change the mode of the pin to GPIO.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Enumeration
	Label	PortPin Direction
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Default	PORT_PIN_IN
	Range	PORT_PIN_IN PORT_PIN_OUT PORT_PIN_HIGH_Z

3.2.2.2.20 PortPinInitialMode

Variable	PortPinInitialMode	
Description	Port pin mode from mode list for use with Port_Init() function. NOTE: This parameter is not used in the current implementation and is retained as per std AUTOSAR_EcucParamDef.arxml file.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Enumeration

	Label	PortPin Initial Mode
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	ReadOnly	true
	Default	PORT_GPIO_MODE
	Range	PORT_ALT0_FUNC_MODE PORT_GPIO_MODE PORT_ALT2_FUNC_MODE PORT_ALT3_FUNC_MODE PORT_ALT4_FUNC_MODE PORT_ALT5_FUNC_MODE PORT_ALT6_FUNC_MODE PORT_ALT7_FUNC_MODE

3.2.2.2.21 PortPinEcucPartitionRef

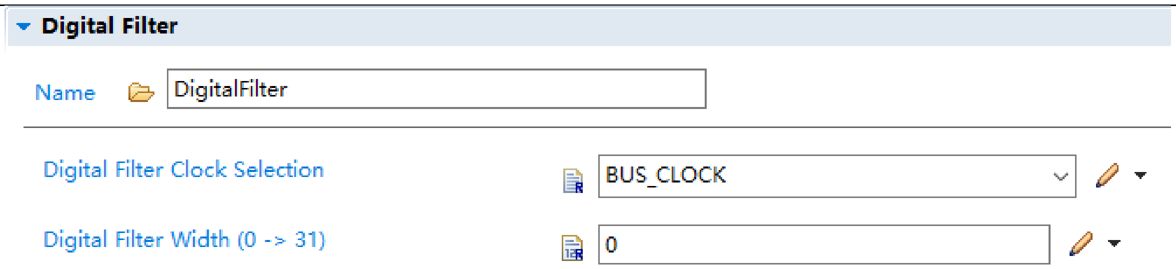
List	PortPinEcucPartitionRef	
Description	Maps the pin to zero or one ECUC partition. The ECUC partitions referenced are a subset of the ECUC partitions where the Port driver is mapped to.	
Screenshot		
Properties	**Property**	**Value**
	Type	List
	Origin	AUTOSAR_ECUC

3.2.2.2.22 PortPinLevelValue

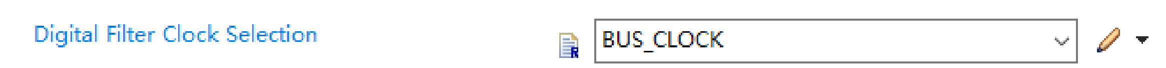
Container	PortPinLevelValue	
Description	Port Pin Level value from Port pin list.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Enumeration
	Label	PortPin Level Value
	Origin	AUTOSAR_ECUC
	SymbolicNameValue	false
	Default	PORT_PIN_LEVEL_LOW

	Range	PORT_PIN_LEVEL_HIGH PORT_PIN_LEVEL_LOW PORT_PIN_LEVEL_NOTCHANGED
	Editable	Only be false in ADC mode.

3.2.2.2.23 DigitalFilter

Container	DigitalFilter	
Description	Configuration of a Digital Filter module available on the platform.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	Digital Filter

3.2.2.2.24 DigitalFilterClock

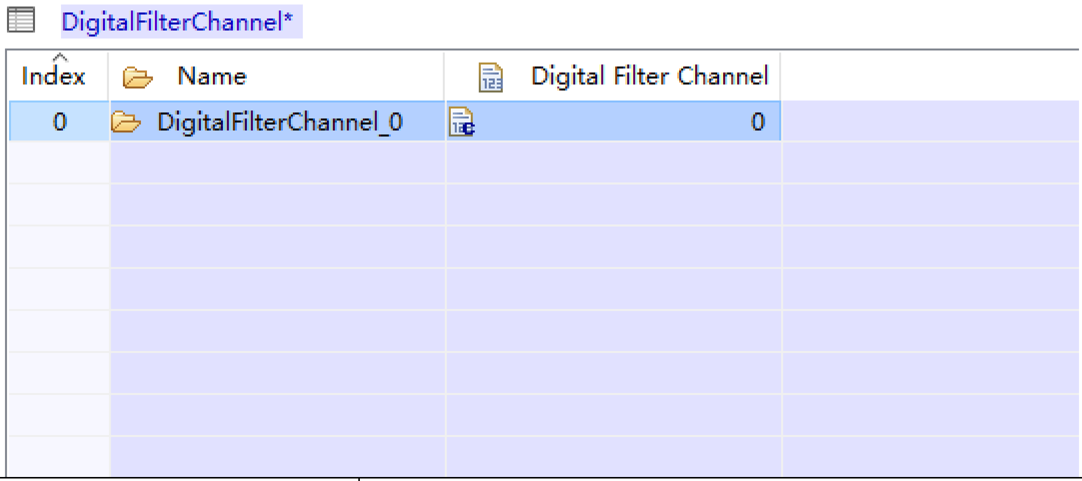
Variable	DigitalFilterClock	
Description	Configures the clock source for the digital input filters.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Enumeration
	Label	Digital Filter Clock Selection
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	BUS_CLOCK
	Range	BUS_CLOCK SIRC128K

3.2.2.2.25 DigitalFilterWidth

Variable	DigitalFilterWidth	
Description	Configures the maximum size of the glitches, in clock cycles, that the digital filter absorbs for the enabled digital filters. Glitches that are longer than this register setting will pass through the digital filter, and glitches that are equal to or less than this register setting are filtered.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	Digital Filter Width
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	0
	Invalid	<=31 >=0

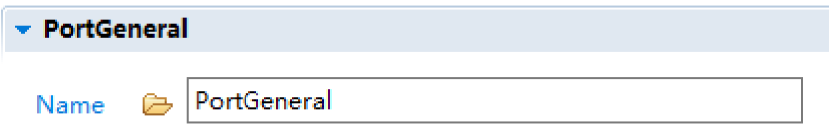
		(the value of the node must be between 0 and 31)

3.2.2.2.26 DigitalFilterChannel

List	DigitalFilterChannel	
Description	Configuration of a Digital Filter channel available on the platform.	
Screenshot		
Properties	**Property**	**Value**
	Type	Map
	Label	Digital Filter Channel

3.2.3 PortConfigurationOfOptApiServices

3.2.3.1 PortGeneral

Container	PortGeneral	
Description	Module wide configuration parameters of the PORT driver.	
Screenshot		
Properties	**Property**	**Value**
	Type	Container

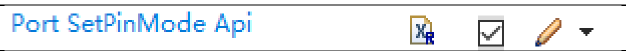
3.2.3.2 PortDevErrorDetect

Variable	PortDevErrorDetect	
Description	Switches the Development Error Detection and Notification ON or OFF. • TRUE: detection and notification are enabled. • FALSE: detection and notification are disabled.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable:Boolean
	Label	Port Development Error Detect
	Origin	AUTOSAR_ECUC
	Default	true

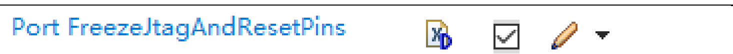
3.2.3.2.1 PortSetPinDirectionApi

Variable	PortSetPinDirectionApi	
Description	Pre-processor switch to enable/disable the use of the function Port_SetPinDirection(). • TRUE: Enabled - Function Port_SetPinDirection() is available. • FALSE: Disabled - Function Port_SetPinDirection() is not available.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	Port SetPinDirection Api
	Origin	AUTOSAR_ECUC
	Default	true

3.2.3.3 PortSetPinModeApi

Variable	PortSetPinModeApi	
Description	Pre-processor switch to enable / disable the use of the function Port_SetPinMode(). • TRUE: Enabled - Function Port_SetPinMode() is available. • FLASE: Disabled - Function Port_SetPinMode() is not available.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	Port SetPinMode Api
	Origin	AUTOSAR_ECUC
	Default	true

3.2.3.4 PortFreezeJtagAndReset

Variable	PortFreezeJtagAndResetPins	
Description	The configuration of JTAG and reset pins(PTA4,PTA5,PTA10，PTC4,PTC5) is unavailable when this switch is ON.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	Port FreezeJtagAndResetPins
	Origin	FLAGCHIP
	Default	true

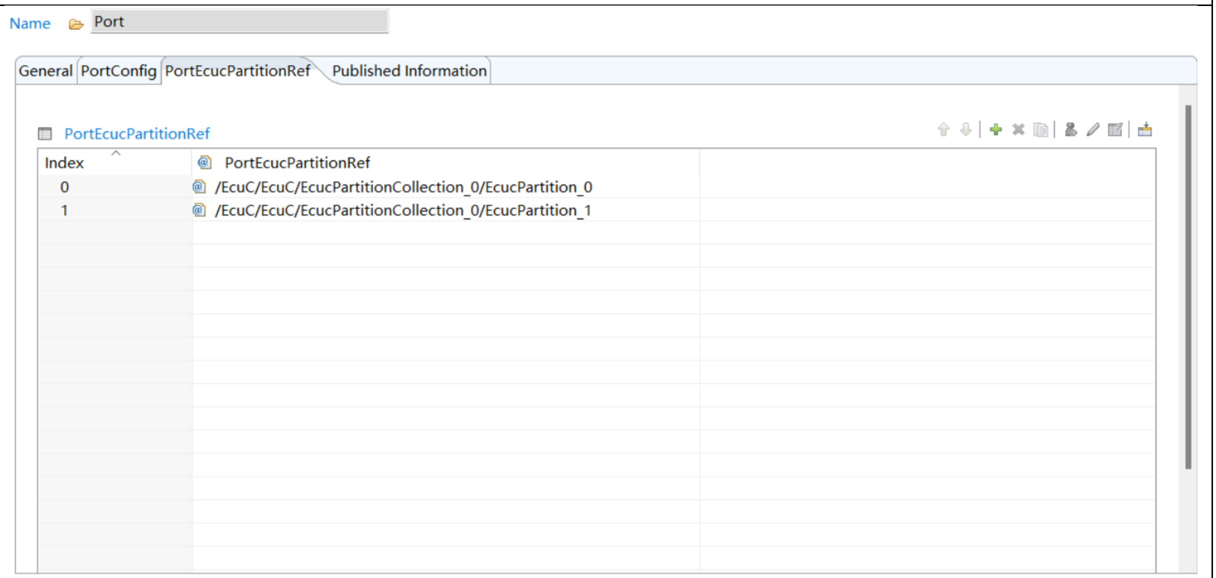
3.2.3.5 PortVersionInfoApi

Variable	PortVersionInfoApi	
Description	Pre-processor switch to enable/disable the API to read out the module's version information. • TRUE: Version info API enabled. • FALSE: Version info API disabled.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Boolean
	Label	Port VersionInfo Api
	Origin	AUTOSAR_ECUC
	Default	true

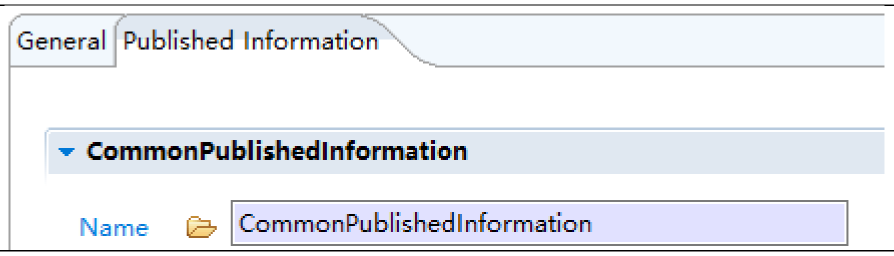
3.2.3.6 PortMulticoreSupport

List	PortMulticoreSupport	
Description	This parameter globally enables the possibility to support multicore. If this parmeter is enabled, at least one EcucPartition needs to be defined (in all variants).	
Screenshot		
Properties	**Property**	**Value**
	Type	BOOLEAN
	Origin	FLAGCHIP
	Default	FALSE

3.2.3.7 PortEcucPartitonRef

List	PortEcucPartitionRef	
Description	Maps the Port driver to zero a multiple ECUC partitions to make the modules API available in this partition.	
Screenshot		
Properties	**Property**	**Value**
	Type	List
	Origin	AUTOSAR_ECUC

3.2.4 CommonPublishedInformation

Container	CommonPublishedInformation
Description	Common container, aggregated by all modules. It contains published information about vendor and versions.
Screenshot	
Properties	N/A

3.2.4.1 ArReleaseMajorVersion

Variable	ArReleaseMajorVersion	
Description	Major version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	false
	Default	4

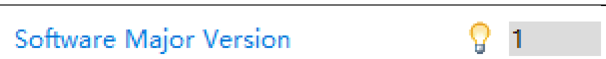
3.2.4.2 ArReleaseMinorVersion

Variable	ArReleaseMinorVersion	
Description	Minor version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	3

3.2.4.3 ArReleaseRevisionVersion

Variable	ArReleaseRevisionVersion	
Description	Revision version number of AUTOSAR specification on which the appropriate implementation is based.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	1

3.2.4.4 SwMajorVersion

Variable	SwMajorVersion	
Description	Major version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	1

3.2.4.5 SwMinorVersion

Variable	SwMinorVersion
Description	Minor version number of the vendor specific implementation of the module. The numbering is vendor

	specific.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	0

3.2.4.6 SwPatchVersion

Variable	SwPatchVersion	
Description	Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	3

3.2.4.7 ModuleId

Variable	ModuleId	
Description	Module ID of this module from Module List.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	124

3.2.4.8 VendorId

Variable	VendorId	
Description	Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.	
Screenshot		
Properties	**Property**	**Value**
	Type	Variable: Integer_Label
	Origin	FLAGCHIP
	SymbolicNameValue	False
	Default	174

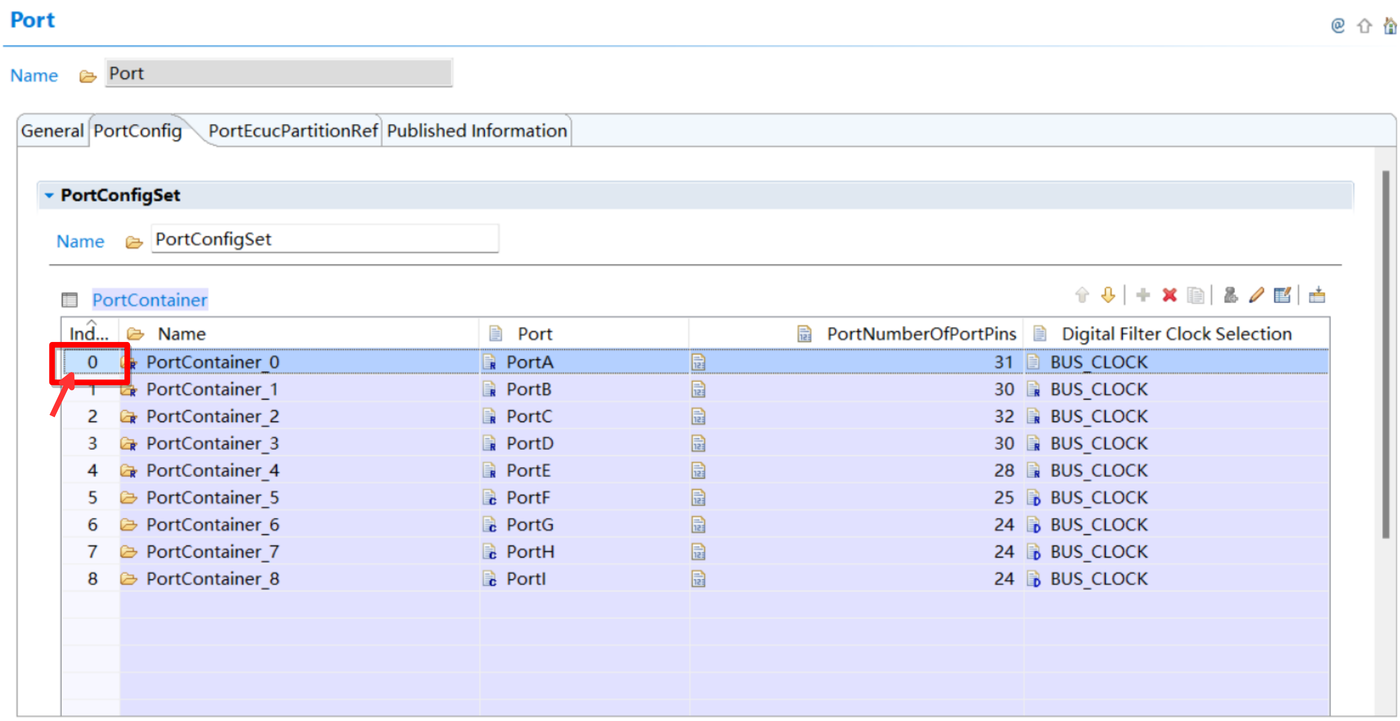
Chapter 4 Configuration Guides

4.1 Port Usage Common Steps

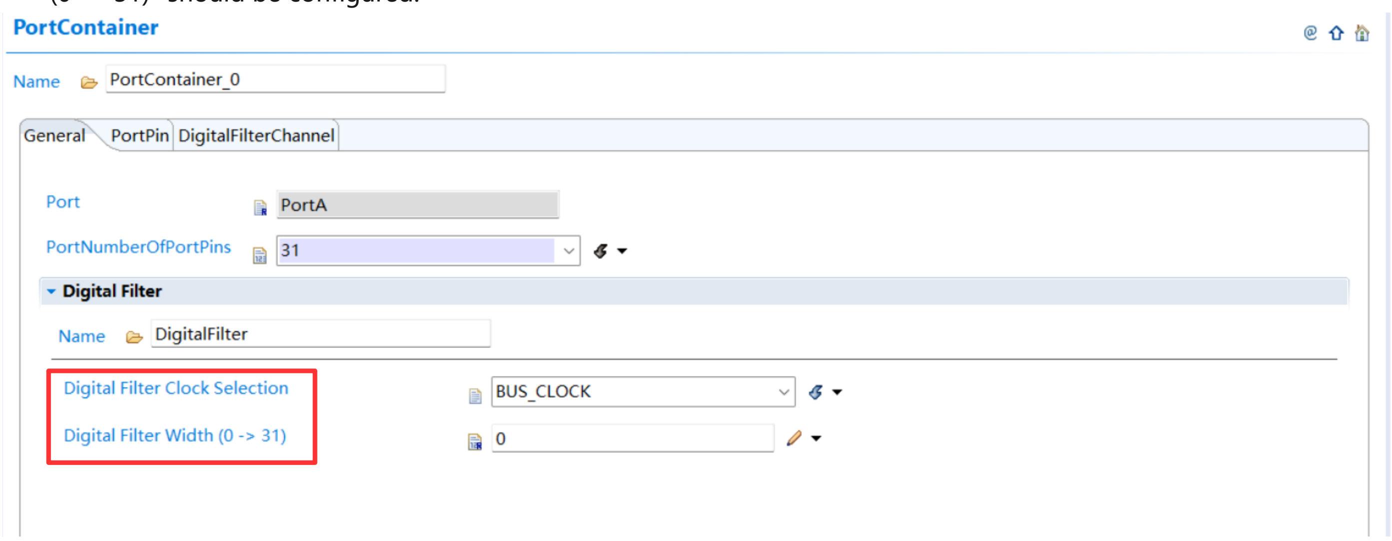
The Port Driver is used for configuring the desired functionality that should be active on several hardware pins. The information about the functionalities available on each of the hardware pins of the platform can be found in the "FC7300 Pinout.xlsx" file attached to the reference manual.

Therefore, the Port module can be configured by following the below 5 steps:

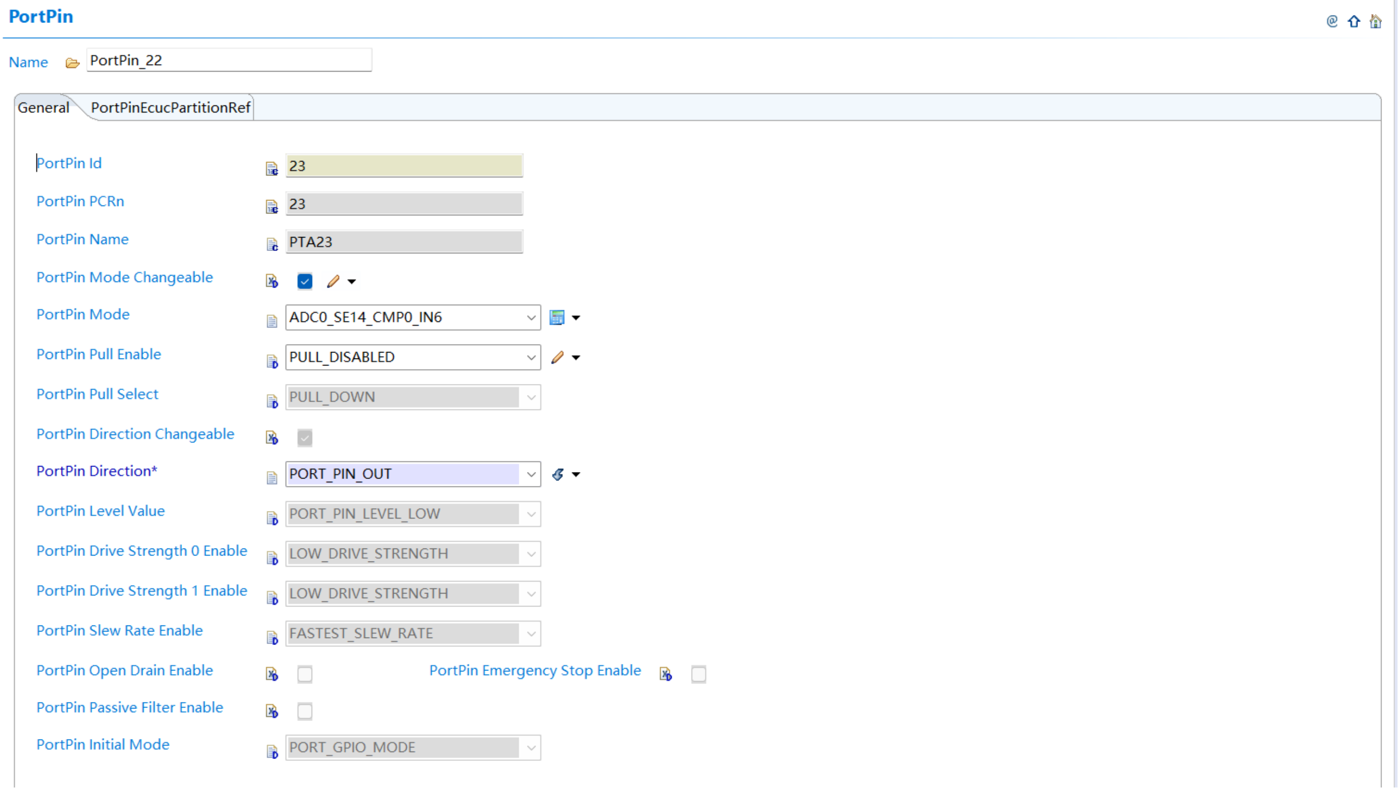
1) Open "FC7300 Pinout.xlsx" file attached to the reference manual, and identify the microcontroller pin you want to use.
2) Go to "PortConfig" tab, find the Port to be configured. Then double-click the "Index" of the Port.



3) If you want to configure "Digital Filter" for the Port, Items "Digital Filter Clock Selection" and "Digital Filter Width (0 -> 31)" should be configured.



4) Go to tab "Port Pin". Locate the Pin you want to configure. Double-click the "Index" of the Pin. Set the configuration item according to your need.

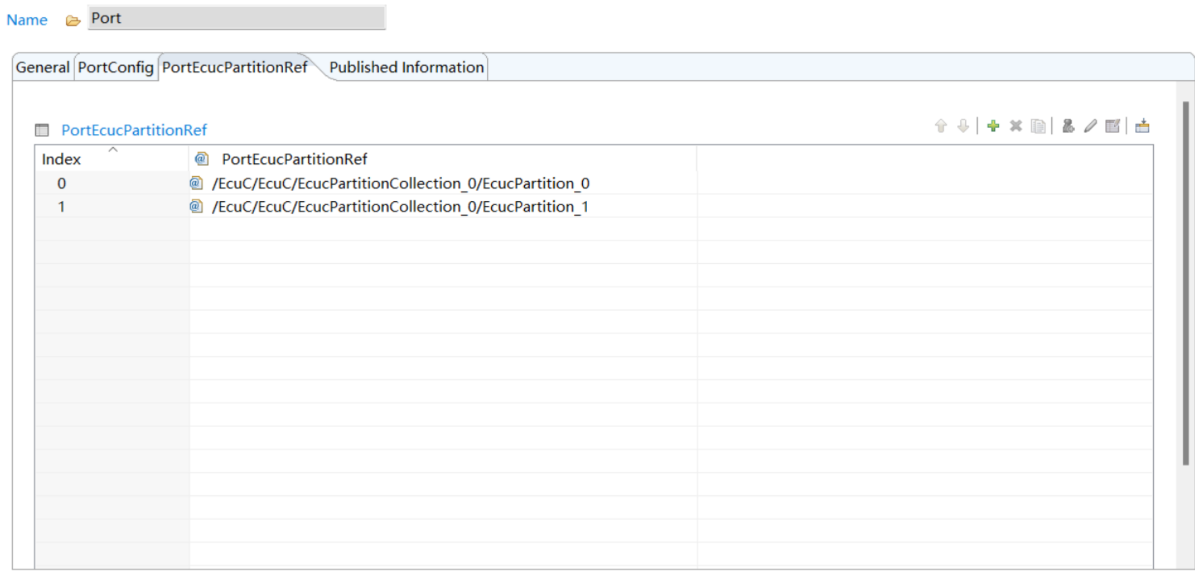


5) Repeat these steps, and make sure all needed Pins have been configured. Right-click the module and select "Generate Configuration" to generate code after configuration is finished.

4.2 Port uses a multi-core step

After the port function is configured, the configuration on the multi-core is mainly composed of the following two steps.

1) Go to "PortEcucPartitionRef" tab, and determine which partitions all pins need to use.



2) Go to "PortPinEcucPartitionRef" tab, and decide which partition to allocate for a particular pin.